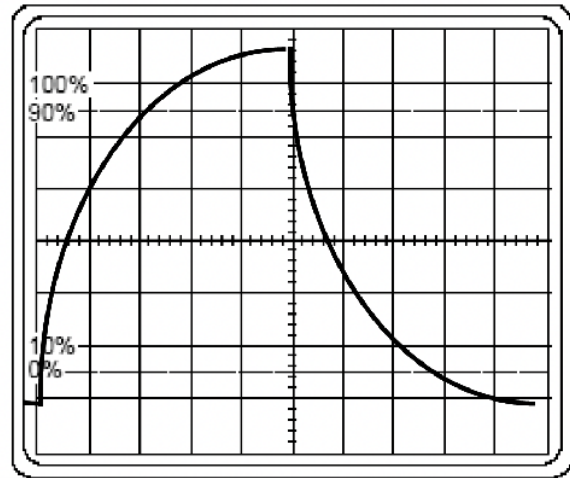


Prelab 7: Fast Decay with RC Circuits

March 27, 2026

1. You have constructed an RC circuit and you see the trace to the right on your oscilloscope screen. You have set the vertical scale to 1 V per square and the horizontal to 20 ms per square. What is the time constant of the RC circuit?



The time constant $\tau = RC$ is equivalent to the “1/e life” of the voltage across the capacitor. That is, at what time t after discharging does the voltage across the capacitor drop to $1/e \approx 36.8\%$ of its initial value?

The range of the voltage is from the line at -10% to the line at 110% . $1/e \approx 36.8\%$ between -10% and 110% is approximately 34.2% , which intersects with the curve about 1 division past the y-axis.

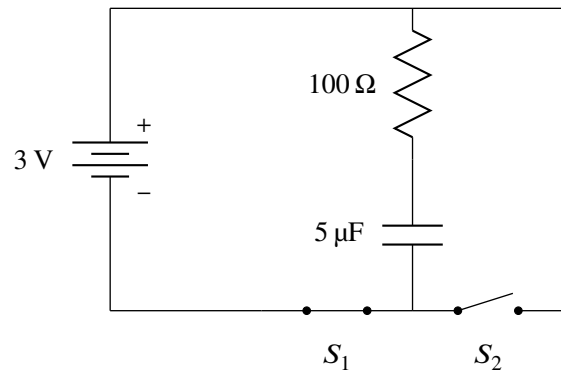
This corresponds to a time constant of approximately:

$$\tau \approx 1 \text{ division} \cdot 20 \text{ ms/division} = 20 \text{ ms.}$$

2. The RC circuit is constructed as shown on the right.
- b. You then change the resistance to 200Ω . How does the time constant change?

The time constant is proportional to the resistance: $\tau \propto R$. Thus, doubling the resistance will double the time constant:

$$\tau' = 2\tau.$$



- b. You then change the capacitance to $20 \mu\text{F}$. How does the time constant change?

The time constant is proportional to the capacitance: $\tau \propto C$. Thus, quadrupling the capacitance will quadruple the time constant:

$$\tau' = 4\tau.$$

- c. You then change to a 9 V battery. How does the time constant change?

The time constant is independent of voltage, so the time constant will not change:

$$\tau' = \tau.$$